CFD 2025-26 – Lab 03

Exercise 1

Let us consider the Stokes problem defined in the unit square $\Omega := (0,1)^2$

$$\begin{cases} -\Delta \boldsymbol{u} + \nabla p = 0, & \text{in } \Omega, \\ \operatorname{div} \boldsymbol{u} = 0, & \text{in } \Omega, \\ \boldsymbol{u} = 1\boldsymbol{i} + 0\boldsymbol{j}, & \text{on } \Gamma^{\text{up}} = \{0 \le x \le 1,\ y = 1\}, \\ \boldsymbol{u} = \boldsymbol{0}, & \text{on } \partial\Omega \backslash \Gamma^{\text{up}}. \end{cases}$$

1. Assemble the monolithic block matrix $\Sigma = \begin{bmatrix} A & B^T \\ -B & 0 \end{bmatrix}$ associated to the problem using the stable pair of spaces $\mathbb{P}^2/\mathbb{P}^1$. Solve the system using the default solver.

2. Now, solve the problem using the GMRES method using a value of 10^{-8} for the tolerance. Does the number of iterations needed to reach the convergence increase with the number of the elements? (Use a uniformly refined grid with $n = 10, 20, 40, 80$ subdivisions). Use the `Firedrake` function `LinearVariationalSolver` with the following parameters:

```
{'ksp_type': 'gmres', 'pc_type': 'none',
 'ksp_rtol': 1.e-8, 'ksp_max_it': 10000}
```

3. Re-do the previous point using an ILU preconditioner.

Exercise 2

Let us consider the Stokes problem defined in the unit square $\Omega := (0,1)^2$

$$\begin{cases} -\Delta \boldsymbol{u} + \nabla p = \boldsymbol{f}, & \text{in } \Omega, \\ \operatorname{div} \boldsymbol{u} = 0, & \text{in } \Omega, \\ \boldsymbol{u} = \boldsymbol{g}, & \text{on } \partial\Omega. \end{cases}$$

1. Compute the functions $\boldsymbol{f}$ and $\boldsymbol{g}$ such that

$$\boldsymbol{u}(x,y) = \begin{bmatrix} -\cos x \sin y \\ \sin x \cos y \end{bmatrix}, \qquad p(x,y) = -\frac{1}{4}(\cos(2x) + \cos(2y)) + \frac{\sin(2)}{4}.$$

2. Write the weak formulation of the differential problem.

3. Using the pairs of finite elements spaces $\mathbb{P}^2/\mathbb{P}^1$ and $\mathbb{P}^1_b/\mathbb{P}^1$ solve the problem with a uniform mesh with an increasing number of subdivisions n. For each pair of spaces fill the following table and verify numerically the convergence rates.

n	N_{dofs}	$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{L^2}$	$\|\nabla \boldsymbol{u} - \nabla \boldsymbol{u}_h\|_{L^2}$	$\|p - p_h\|_{L^2}$
5				
10				
20				
40				

4 Let us introduce the following adjusted pressure:

$$\widehat{p}_h(x, y) = p_h(x, y) + \frac{1}{|\Omega|} \int_\Omega (p - p_h) d\boldsymbol{x}.$$

Repeat the convergence tests of the previous point, considering $\|p - \widehat{p}_h\|_{L^2}$ instead of $\|p - p_h\|_{L^2}$. What can you observe? Can you tell why?